



5CM512

Communication and Security Protocols

# Potential Projects: KEY DETAILS

# SECURE CHAT APPLICATION

## Objectives

- Develop a messaging app that ensures secure communication between users.

## Description

- Implement end-to-end encryption using protocols like Diffie-Hellman key exchange or AES for message encryption.
- Use a protocol like TLS to secure data in transit.

## Learning Outcomes

- Understand encryption methods.
- Explore how secure communication channels are established.

# IoT Device Communication Security

## Objectives

- Simulate or implement secure communication between IoT devices.

## Description

- Use MQTT or CoAP as a communication protocol.
- Implement data encryption and authentication mechanisms like OAuth 2.0 or TLS.
- Address specific IoT challenges such as lightweight encryption and energy efficiency.

## Learning Outcomes

- Understand the security concerns in IoT networks.
- Learn lightweight encryption and resource management in constrained devices.

# Man-in-the-Middle Attack Simulation

## Objectives

- Demonstrate a Man-in-the-Middle (MITM) attack and implement countermeasures.

## Description

- Create a scenario where data interception occurs over an insecure communication protocol (e.g., HTTP).
- Highlight the vulnerabilities exploited in MITM.
- Implement solutions like HTTPS, certificate pinning, or using secure DNS.

## Learning Outcomes

- Understand vulnerabilities in communication protocols.
- Learn practical mitigation strategies.

# Implementing Blockchain-Based Secure Communication

## Objectives

- Build a prototype for secure communication using blockchain.

## Description

- Use blockchain for key distribution or message authentication.
- Explore smart contracts for automating certain secure communication tasks.

## Learning Outcomes

- Learn how distributed systems can enhance security.
- Explore blockchain concepts like consensus and immutability in communication.

# Network Traffic Analysis Tool

## Objectives

- Build a tool to analyze and identify malicious activity in network traffic.

## Description

- Use Wireshark or a similar framework for capturing packets.
- Implement functionality to detect anomalies like packet injections or unusual patterns.

## Learning Outcomes

- Learn about packet-level communication.
- Understand traffic analysis and anomaly detection.

# Implementing HTTPS with Self-Signed Certificates

## Objectives

- Build a small web application that uses HTTPS.

## Description

- Set up a web server and configure it with a self-signed SSL/TLS certificate.
- Demonstrate how HTTPS secures communication and prevents eavesdropping.
- Discuss the role of Certificate Authorities (CAs).

## Learning Outcomes

- Understand the role of TLS/SSL in secure communication.
- Learn about the Public Key Infrastructure (PKI).



# Secure File Transfer Protocol

## Objectives

- Create a secure file-sharing application.

## Description

- Implement protocols like SFTP or FTPS for secure file transfers.
- Ensure file integrity using hashing algorithms like SHA-256. Add role-based access control for users.

## Learning Outcomes

- Learn secure file transfer protocols.
- Explore authentication and integrity checks.



# Simulation of Distributed Denial of Service (DDoS) Attack

## Objectives

- Simulate a DDoS attack and implement mitigation techniques.

## Description

- Create a controlled environment to simulate a DDoS attack on a server.
- Analyze the attack patterns and implement countermeasures like rate-limiting or using a Content Delivery Network (CDN).

## Learning Outcomes

- Understand DDoS attack mechanisms.
- Learn about detection and mitigation strategies.

# Secure Voting System

## Objectives

- Design a secure digital voting system.

## Description

- Implement cryptographic techniques to ensure voter privacy and ballot integrity.
- Use protocols like Zero-Knowledge Proofs or Homomorphic Encryption.

## Learning Outcomes

- Explore the challenges of secure voting systems.
- Apply advanced cryptographic techniques.

# Secure Voting System

## Objectives

- Assess and secure a wireless network.

## Description

- Perform penetration testing on a Wi-Fi network using tools like Aircrack-ng.
- Implement security measures like WPA3, MAC filtering, and hidden SSIDs.

## Learning Outcomes

- Understand vulnerabilities in wireless networks.
- Learn practical ways to secure wireless communication.

# Any Questions?

